Design and Modelling of Fins of Chevron Actuator for Optical Switching

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***Abstract*—Chevron (V-beam) thermal actuators are widely employed in microelectromechanical systems (MEMS) for precision positioning applications, including optical switching in fiber-optic networks. In such systems, the actuator functions as a linear motor to control the three-dimensional position of micromirrors, enabling dynamic routing of optical signals. The switching speed of these devices is fundamentally limited by the coupled thermal and mechanical dynamics of the actuator, with thermal transport often acting as the dominant bottleneck.**

In this work, we present a comprehensive analysis of the frequency response of chevron actuators, with a focus on improving high-frequency performance. We first investigate the influence of input excitation waveforms on thermal actuation efficiency and demonstrate that square-wave inputs maximize displacement at higher frequencies due to enhanced energy delivery and thermal retention. The actuator is then modeled as decoupled thermal and mechanical systems, revealing that thermal dynamics impose a low-pass constraint that suppresses mechanical resonance in typical designs.

To address this limitation, we propose the incorporation of fin-like structures to enhance heat transfer, thereby increasing the thermal bandwidth of the system. A design map is developed to delineate thermal-limited and mechanical-limited operating regimes, providing a systematic framework for actuator optimization. The results highlight the critical role of thermal management in enabling high-speed operation and offer design guidelines for next-generation MEMS optical switches.

***Index Terms*—Chevron actuator, MEMS thermal actuator, optical switching, thermo-mechanical modeling, frequency response, thermal time constant, mechanical resonance, high-speed actuation, waveform optimization, thermal management**

I. Introduction

MICROELECTROMECHANICAL systems (MEMS) have enabled compact, low-power, and high-precision actuation mechanisms for a wide range of applications, including optical communication, sensing, and micro-robotics [1]. Among these, chevron (V-beam) thermal actuators have emerged as a robust solution for generating controlled linear motion through Joule heating-induced thermal expansion [2]. In optical switching systems, these actuators are used to precisely position micromirrors, enabling dynamic routing of light in fiber-optic networks.

With the increasing demand for higher data rates and faster switching speeds in optical communication systems, there is a growing need to improve the dynamic performance of MEMS actuators. The maximum achievable switching frequency of chevron actuators is constrained by their underlying thermal and mechanical dynamics. While the mechanical subsystem is capable of high-frequency operation due to its low mass and high stiffness, the thermal subsystem exhibits a significantly slower response governed by heat diffusion and dissipation [3], [4]. As a result, most practical chevron actuators operate in a thermally limited regime, preventing them from reaching their mechanical resonance and limiting overall actuation speed.

Understanding and overcoming this limitation is critical for the design of high-speed MEMS optical switches. Prior work has largely focused on geometric optimization and material selection [**?**], [5]; however, the role of input excitation and thermal management in determining actuator bandwidth remains underexplored. In particular, the influence of waveform characteristics on energy delivery and the potential of enhanced heat dissipation mechanisms, such as fin-like structures, warrant systematic investigation [**?**].

In this work, we present a unified framework for analyzing and optimizing the frequency response of chevron actuators. We begin by examining the effect of different input waveforms on thermal actuation efficiency, followed by a decoupled analysis of thermal and mechanical subsystems. This leads to the formulation of a combined thermo-mechanical response model, which highlights the dominance of thermal limitations. Building on this understanding, we derive a design condition that determines whether the actuator operates in a thermal-limited or mechanical-limited regime and visualize it using a design map. Finally, we propose the use of fin-enhanced geometries to improve heat transfer and extend the operational bandwidth of the actuator.

The contributions of this work are threefold:

- A systematic analysis of waveform-dependent thermal actuation efficiency.
- A unified thermo-mechanical modeling framework for frequency response characterization.
- A design methodology for transitioning from thermal-limited to mechanically accessible regimes through enhanced heat dissipation.

These insights provide a pathway toward the design of high-speed MEMS actuators for next-generation optical switching applications.

II. Analysis of Chevron Actuator Dynamics

A. Effect of Input Waveform on Thermal Actuation Efficiency

The chevron actuator is fundamentally driven by Joule heating, where the input electrical signal determines the temporal evolution of temperature and hence displacement. The thermal dynamics are governed by the first-order energy balance:

$$\rho c V \frac{dT}{dt} = P(t) - hA(T - T_{amb}) \tag{1}$$

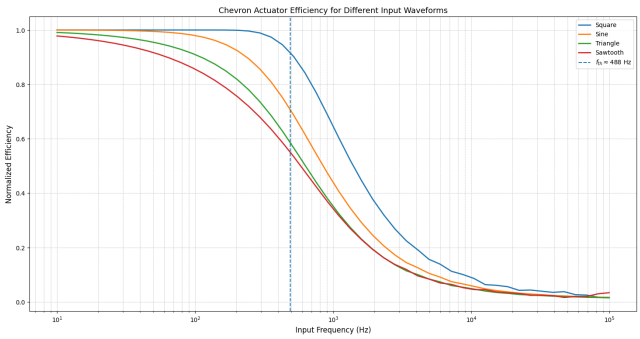

Fig. 1. Normalized actuator efficiency for different input waveforms.

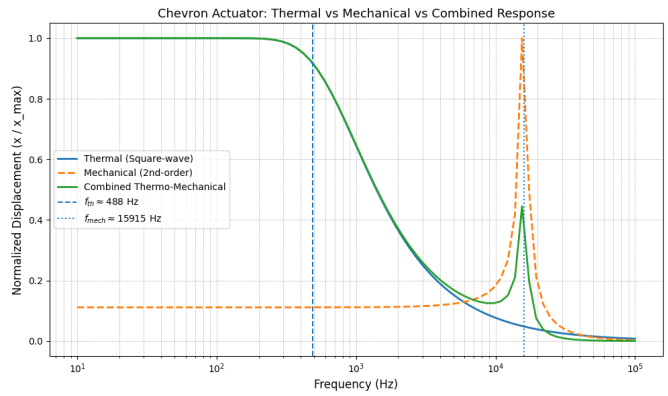

Fig. 2. Thermal, mechanical, and combined frequency response of the actuator.

where ρ, c, and V represent density, specific heat, and volume respectively, while hA denotes the effective heat dissipation coefficient.

Defining $\theta = T - T_{amb}$, the system reduces to:

$$\frac{d\theta}{dt} + \frac{1}{\tau_{th}}\theta = \frac{P(t)}{\rho c V} \tag{2}$$

where the thermal time constant is:

$$\tau_{th} = \frac{\rho c V}{hA} \tag{3}$$

The actuator displacement is proportional to the temperature rise:

$$x(t) \propto \Delta T(t) \tag{4}$$

To evaluate performance, we define the normalized efficiency as:

$$\eta(f) = \frac{x_{\text{amp}}(f)}{x_{\text{ref}}} \tag{5}$$

where x_{ref} corresponds to the steady-state displacement under equivalent average power.

Different input waveforms (square, sinusoidal, triangular, etc.) were analyzed. It was observed that square-wave excitation provides superior efficiency at higher frequencies due to its ability to deliver peak power instantaneously and exploit thermal memory effects.

B. Decoupled Thermal and Mechanical Modeling

The chevron actuator can be decomposed into thermal and mechanical subsystems.

1) Thermal Model: The thermal system behaves as a first-order low-pass system with transfer function:

$$H_{th}(j\omega) = \frac{1}{1 + j\omega\tau_{th}} \tag{6}$$

The corresponding frequency response magnitude is:

$$|H_{th}(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau_{th})^2}} \tag{7}$$

This limits the maximum effective operating frequency:

$$f_{th} = \frac{1}{2\pi\tau_{th}} \tag{8}$$

2) Mechanical Model: The mechanical behavior is modeled as a second-order mass-spring-damper system:

$$m\ddot{x} + c\dot{x} + kx = F(t) \tag{9}$$

The frequency response is:

$$|H_{mech}(j\omega)| = \frac{1}{\sqrt{(1 - (\omega/\omega_n)^2)^2 + (2\zeta\omega/\omega_n)^2}} \tag{10}$$

where:

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{2\sqrt{km}} \tag{11}$$

The mechanical resonance frequency is:

$$f_{mech} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{12}$$

3) Combined Thermo-Mechanical Response: The overall actuator response can be approximated as:

$$H_{total}(j\omega) = H_{th}(j\omega) \cdot H_{mech}(j\omega) \tag{13}$$

It is observed that in most practical scenarios:

$$f_{th} \ll f_{mech} \tag{14}$$

which implies that thermal dynamics dominate and suppress the mechanical resonance.

C. Design Map for Thermal vs Mechanical Limitation

To determine the dominant limitation regime, we equate thermal and mechanical bandwidths:

$$f_{th} = f_{mech} \tag{15}$$

Substituting the respective expressions:

$$\frac{hA}{2\pi\rho c V} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{16}$$

Simplifying:

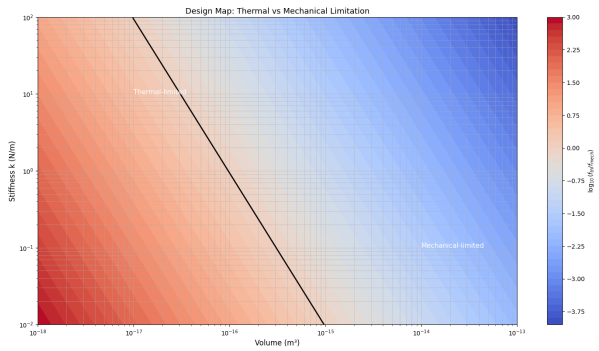

Fig. 3. Design map illustrating thermal and mechanical limitation regimes.

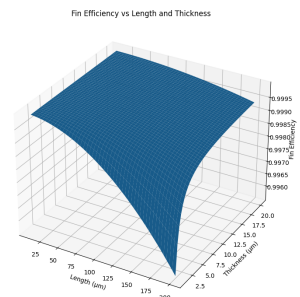

Fig. 4. Fin efficiency as a function of fin length and thickness.

$$\frac{hA}{\rho c V} = \sqrt{\frac{k}{m}} \tag{17}$$

Squaring both sides:

$$\boxed{\frac{(hA)^2}{(\rho c V)^2} = \frac{k}{m}} \tag{18}$$

This relation defines the boundary between thermal-limited and mechanical-limited regimes.

A dimensionless ratio is defined as:

$$R = \frac{f_{th}}{f_{mech}} = \frac{hA}{\rho c V} \cdot \sqrt{\frac{m}{k}} \tag{19}$$

- $R < 1$: Thermal-limited regime
- $R > 1$: Mechanical-limited regime
- $R = 1$: Boundary condition

D. Fin Efficiency Characterization

To quantify the effectiveness of fin structures, we define fin efficiency as the ratio of actual heat transfer to the ideal heat transfer if the entire fin were at base temperature:

$$\eta_f = \frac{q_{\text{actual}}}{q_{\text{ideal}}} \tag{20}$$

For a straight fin with uniform cross-section, the efficiency can be expressed as:

$$\eta_f = \frac{\tanh(mL)}{mL} \tag{21}$$

where:

$$m = \sqrt{\frac{hP}{kA_c}} \tag{22}$$

Here, P is the perimeter, A_c is the cross-sectional area, and k is the thermal conductivity.

A parametric study was conducted by varying fin length and thickness. The resulting efficiency surface is shown in Fig. 4.

The results show that fin efficiency remains close to unity across the explored design space, indicating that the fins operate near ideal conditions. However, efficiency decreases slightly with increasing fin length due to temperature gradients along the fin. Increasing thickness improves efficiency by reducing thermal resistance.

These observations suggest that fin design should prioritize increased surface area for heat transfer while maintaining high efficiency, ensuring that the added structures effectively contribute to thermal bandwidth enhancement without significant losses.

E. Design Trade-off Between Efficiency and Heat Dissipation

To enhance the thermal performance of chevron actuators, fin-like extensions can be incorporated to improve heat dissipation. However, this introduces a fundamental trade-off between thermal bandwidth and actuation efficiency. Increasing the effective heat transfer coefficient hA improves thermal response speed but reduces steady-state temperature rise, thereby lowering displacement.

To capture this trade-off, we define normalized efficiency as:

$$\eta = \frac{\Delta T_{\text{amp}}}{\Delta T_{\text{ref}}} \tag{23}$$

where ΔT_{ref} corresponds to the temperature rise under nominal conditions without enhanced heat dissipation.

The thermal bandwidth is given by:

$$f_{th} = \frac{hA}{2\pi \rho c V} \tag{24}$$

Thus, increasing hA improves f_{th} but simultaneously reduces ΔT:

$$\Delta T \propto \frac{P}{hA} \tag{25}$$

A parametric sweep over geometric variables (length and thickness of fins) was performed to evaluate this trade-off. Fig. 5 shows contour plots of normalized efficiency (solid lines) and heat dissipation (dashed lines).

The results indicate that increasing fin thickness significantly enhances heat dissipation, while increasing fin length provides diminishing returns beyond a certain threshold. Importantly, regions of high heat dissipation correspond to reduced actuation efficiency, highlighting the need for optimal geometric selection. This trade-off forms the basis for designing fins that improve frequency response without excessively compromising displacement.

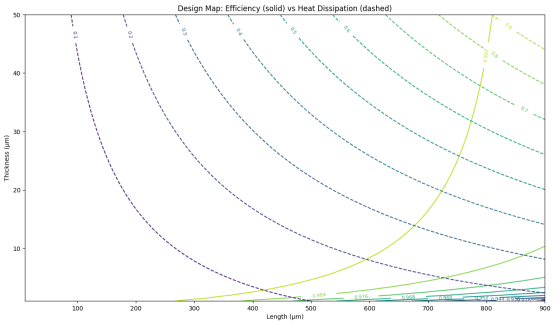

Fig. 5. Design map showing trade-off between normalized efficiency (solid contours) and heat dissipation (dashed contours) as a function of fin geometry.

F. Key Observations

- Square-wave excitation maximizes thermal efficiency at high frequencies.
- Thermal dynamics impose a fundamental bandwidth limitation on actuation.
- Mechanical resonance is typically not excited due to insufficient thermal bandwidth.
- The derived design map provides a systematic framework for actuator optimization.
- Enhanced heat dissipation through fin structures increases thermal bandwidth but introduces a trade-off with steady-state displacement.
- Fin efficiency remains close to unity across practical geometries, indicating that performance improvements are primarily governed by geometric optimization of surface area rather than intrinsic fin losses.

III. Conclusion

In this work, we presented a comprehensive analysis of the frequency response of chevron thermal actuators, with a focus on their application in high-speed optical switching. By investigating the effect of input waveforms, we demonstrated that square-wave excitation improves thermal actuation efficiency at higher frequencies. A decoupled thermo-mechanical modeling framework was developed, revealing that thermal dynamics impose the dominant limitation on actuator bandwidth and suppress mechanical resonance in typical designs. Furthermore, a design map was formulated to identify the conditions under which the actuator operates in thermal-limited or mechanical-limited regimes.

These findings highlight the critical role of thermal management in enhancing actuator performance. As an extension of this work, the design and analysis of fin-based geometries for improved heat transfer are currently underway, with the objective of increasing thermal bandwidth and enabling higher-frequency operation.

All the code for simulations and the results obtained can be found at https://github.com/simply-sankalp/chevron-simulation

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